Band-1 Computational Experiment: Chromatic number of the plane — unit-distance graph hunt

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note does **not** claim that $\chi(\mathbb{R}^2)$ is solved. It does **not** invent a fake 5-chromatic unit-distance graph. Standing bounds remain

$$\boxed{5 \leq \chi(\mathbb{R}^2) \leq 7}.$$

Tonight's probe **code-certifies** small explicit unit-distance graphs (triangle $\chi = 3$, Moser spindle $\chi = 4$) and **independently re-certifies** a published Heule 553-vertex mosaic as a finite unit-distance graph with $\chi = 5$ (vtx→edges match + SAT), which implies the literature lower bound $\chi(\mathbb{R}^2) \geq 5$ already due to de Grey (2018). **No movement of the standing 5–7 window.**

Abstract

The Hadwiger–Nelson problem asks for $\chi(\mathbb{R}^2)$, the minimum number of colors so that no two points at Euclidean distance 1 share a color. Equivalently: the chromatic number of the infinite unit-distance graph on $\mathbb{R}^2$.

We implement a searchable unit-distance pipeline (coordinates → unit edges → coloring backtrack / SAT) and run it on:

1. equilateral triangle — box $\chi = 3$;
2. Moser spindle (1961) — box $\chi = 4$ (not 3-colorable; 4-colorable);
3. Heule 553-vertex mosaic (2018) — box: published `.vtx` rebuilds published 2722 unit edges; Heule SBP CNF is UNSAT under Cadical (not 4-colorable); our 5-color CNF is SAT with a verified proper witness ⇒ finite $\chi = 5$.

Literature ledger locks de Grey (2018) for $\chi \geq 5$ and the classical hexagonal tiling for $\chi \leq 7$.

Problem statement

Open target. Determine $\chi(\mathbb{R}^2) \in \{5, 6, 7\}$.

Finite method. Any finite unit-distance graph $G \subset \mathbb{R}^2$ satisfies $\chi(G) \leq \chi(\mathbb{R}^2)$. Hence a k-chromatic finite unit-distance graph proves $\chi(\mathbb{R}^2) \geq k$. Upper bounds come from explicit colorings of the whole plane.

Literature ledger (standing 5–7)

Bound	Value	Source
Lower (classical)	$\chi \geq 4$	Moser & Moser, *Canad. Math. Bull.* **4** (1961) 187–189 (spindle)
Lower (current)	$\chi \geq 5$	de Grey, *Geombinatorics* **28** (2018) 18–31; arXiv:1804.02385 (orig. shrink to $ $1581 verts)
Smaller mosaics	$\chi \geq 5$	Heule, *Geombinatorics* **28** (2018) 32–50; arXiv:1805.12181 (553 verts); later 510 / Parts 509
Upper	$\chi \leq 7$	Classical hexagonal lattice / tiling 7-coloring (Isbell; standard)

Standing enclosure adopted here (2026-09): $5 \leq \chi(\mathbb{R}^2) \leq 7$.

What would raise 5 or lower 7

- **Raise the lower bound above 5:** exhibit a finite unit-distance graph with no 5-coloring ($\chi \geq 6$), or a proof that every 5-coloring of the plane forces a monochromatic unit segment.
- **Lower the upper bound below 7:** exhibit a 6-coloring of $\mathbb{R}^2$ with no monochromatic unit distance (measurable/tiling constructions are the usual route), or prove $\chi \leq 6$.

This run does **neither**.

Method

Pipeline

1. **Coordinates.** Algebraic Moser generators; Mathematica-style `.vtx` parse (sympy) for Heule/Moser files from `marijnheule/CNP-SAT` and `vasnesterov/HadwigerNelson`.
2. **Unit edges.** Pairwise distances; keep pairs with $||p_i - p_j| - 1| \leq$ `tol` (10^{-9} Moser; 10^{-8} Heule float rebuild).
3. **Small-χ certification.** Sequential coloring backtrack with v_0 fixed (triangle, Moser).
4. **Large-χ certification.** DIMACS edge match; Cadical195 on Heule `553-4-sbp.cnf` (UNSAT $\Rightarrow$ not 4-colorable); our standard k-color CNF for $k = 5$ (SAT + proper-witness check).

Method limits

- Float rebuild of Heule edges uses tolerance 10^{-8} (not formal interval arithmetic / Lean `build_edge`).
- 4-color UNSAT uses Heule's published SBP CNF (symmetry-breaking predicates), not a from-scratch DRAT proof check tonight.
- Golomb 10-vertex $\chi = 4$ embedding was **not** finalized in this wrap (tentative constructions gave $\chi = 3$); Moser covers the classical $\chi \geq 4$ rung.
- No claim that $\chi(\mathbb{R}^2) \in \{5, 6, 7\}$ is resolved to a single value.

Results (numbers!)

Box-certified graphs

Graph	n	m	Box χ	Method
Equilateral triangle	3	3	**3**	backtrack
Moser spindle	7	11	**4**	algebraic + backtrack; not 3-colorable
Moser from `moser.vtx`	7	11	**4**	vtx rebuild + backtrack
Heule 553	553	2722	**5**	vtx↔edge match; SBP UNSAT; 5-color SAT witness

Moser spindle (detail)

- Witness 4-coloring: `[0, 1, 2, 1, 2, 0, 3]`
- Degrees: `[4, 3, 3, 3, 3, 3, 3]`
- Backtrack nodes to reject 3 colors: 17; accept 4 colors: 7 nodes.

Heule 553 (detail)

- `.vtx` parse ≈ 0.769 s; unit-edge rebuild ≈ 0.014 s; **match published `553.edge` = True**.
- Cadical on `553-4-sbp.cnf`: SAT=False in ≈ 1.951 s (2212 vars, 11444 clauses).
- Our 5-color CNF: SAT=True in ≈ 0.011 s; witness proper=True.

Code-certified vs literature-cited

Claim	Status tonight
$\chi(\triangle) = 3$	**code-certified**
Moser spindle $\chi = 4$ $\Rightarrow \chi(\mathbb{R}^2) \geq 4$	**code-certified**
Heule 553 finite $\chi = 5$ $\Rightarrow \chi(\mathbb{R}^2) \geq 5$	**code-certified** (published mosaic + box SAT/rebuild)
de Grey 2018 first $\chi \geq 5$ breakthrough	**literature-locked** (citation; 1581 not re-SAT'd as primary)
$\chi(\mathbb{R}^2) \leq 7$ hexagonal tiling	**literature-locked**
Exact value of $\chi(\mathbb{R}^2)$	**open** (still 5, 6, or 7)

Reproducibility

- Code / data: `/workspace/experiments/band1/chromatic_plane/`
- Key modules: `vtx_parse.py`, `unit_distance.py`, `coloring.py`, `graphs_lib.py`

- Data: `data/moser.vtx`, `data/553.vtx`, `data/553.edge`, `data/553-4-sbp.cnf` (from Heule CNP-SAT / HadwigerNelson)
- Results: `results.json`
- Log: `/workspace/experiments/logs/2026-09-11_chromatic_plane.log`

Conclusion

Box-certified: triangle $\chi = 3$; Moser spindle $\chi = 4$; Heule 553 finite $\chi = 5$ (unit-distance rebuild match + non-4-colorability + 5-colorability). **Literature standing plane bound unchanged:** $5 \leq \chi(\mathbb{R}^2) \leq 7$. Raising 5 or lowering 7 remains future work. $\chi(\mathbb{R}^2)$ **not solved.**